

$$u_{11} u_{12} u_{13} u_{14} \rightarrow$$

$$\begin{aligned} & \frac{1}{16\pi^2} \left(\frac{1}{216} \frac{1}{m_e^2} (7 g_1^2 c_Y^2 \overline{y_u}^{p13} y_u^{p14} \delta_{i1i2} + 2 s_Y^2 \overline{y_e}^{i2p} y_e^{i1p} (-27 c_Y^2 \overline{y_u}^{ri3} y_u^{ri4} + 7 g_1^2 \delta_{i3i4})) - \right. \\ & \frac{4}{81} \sum_p g_1^4 \text{LF}_{3,0} [m_d^p] \delta_{i1i2} \delta_{i3i4} + \frac{5}{54} \sum_p g_1^4 \text{LF}_{4,-1} [m_d^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{16}{405} \sum_p g_1^4 \text{LF}_{5,-2} [m_d^p] \delta_{i1i2} \delta_{i3i4} - \frac{4}{27} \sum_p g_1^4 \text{LF}_{3,0} [m_e^p] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{5}{18} \sum_p g_1^4 \text{LF}_{4,-1} [m_e^p] \delta_{i1i2} \delta_{i3i4} - \frac{16}{135} \sum_p g_1^4 \text{LF}_{5,-2} [m_e^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{2}{27} \sum_p g_1^4 \text{LF}_{3,0} [m_l^p] \delta_{i1i2} \delta_{i3i4} + \frac{5}{36} \sum_p g_1^4 \text{LF}_{4,-1} [m_l^p] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{8}{135} \sum_p g_1^4 \text{LF}_{5,-2} [m_l^p] \delta_{i1i2} \delta_{i3i4} - \frac{2}{81} \sum_p g_1^4 \text{LF}_{3,0} [m_q^p] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{5}{108} \sum_p g_1^4 \text{LF}_{4,-1} [m_q^p] \delta_{i1i2} \delta_{i3i4} - \frac{8}{405} \sum_p g_1^4 \text{LF}_{5,-2} [m_q^p] \delta_{i1i2} \delta_{i3i4} - \frac{16}{81} \sum_p g_1^4 \text{LF}_{3,0} [m_u^p] \\ & \delta_{i1i2} \delta_{i3i4} + \frac{10}{27} \sum_p g_1^4 \text{LF}_{4,-1} [m_u^p] \delta_{i1i2} \delta_{i3i4} - \frac{64}{405} \sum_p g_1^4 \text{LF}_{5,-2} [m_u^p] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{18} (g_1^2 c_Y^2 \overline{y_u}^{p13} y_u^{p14} \delta_{i1i2} + s_Y^2 \overline{y_e}^{i2p} y_e^{i1p} (-9 c_Y^2 \overline{y_u}^{ri3} y_u^{ri4} + 4 g_1^2 \delta_{i3i4})) \text{LF}_{1,2} [m_\oplus] + \\ & \frac{1}{36} (3 g_1^2 c_Y^2 \overline{y_u}^{p13} y_u^{p14} \delta_{i1i2} + s_Y^2 \overline{y_e}^{i2p} y_e^{i1p} (9 c_Y^2 \overline{y_u}^{ri3} y_u^{ri4} - 2 g_1^2 \delta_{i3i4})) \text{LF}_{2,1} [m_\oplus] + \\ & \frac{1}{108} g_1^2 (6 s_Y^2 \overline{y_e}^{i2p} y_e^{i1p} \delta_{i3i4} - (9 c_Y^2 \overline{y_u}^{p13} y_u^{p14} + 8 g_1^2 \delta_{i3i4}) \delta_{i1i2}) \text{LF}_{3,0} [m_\oplus] + \\ & \frac{5}{36} g_1^4 \text{LF}_{4,-1} [m_\oplus] \delta_{i1i2} \delta_{i3i4} - \frac{8}{135} g_1^4 \text{LF}_{5,-2} [m_\oplus] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{2}{27} g_1^4 \text{LF}_{3,0} [\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{9} g_1^4 \text{LF}_{4,-1} [\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \frac{16}{135} g_1^4 \text{LF}_{5,-2} [\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{72} g_1^4 \text{LF}_{2,1,0} [m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{72} g_1^4 \text{LF}_{2,2,-1} [m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{36} g_1^4 \text{LF}_{3,1,-1} [m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{72} g_1^4 \text{LF}_{4,1,-2} [m_1, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{72} g_1^4 \text{LF}_{2,1,0} [m_1, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{72} g_1^4 \text{LF}_{2,2,-1} [m_1, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{36} g_1^4 \text{LF}_{3,1,-1} [m_1, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{72} g_1^4 \text{LF}_{4,1,-2} [m_1, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{2}{81} g_1^4 \text{LF}_{2,1,0} [m_1, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{2}{81} g_1^4 \text{LF}_{2,2,-1} [m_1, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{4}{81} g_1^4 \text{LF}_{3,1,-1} [m_1, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{2}{81} g_1^4 \text{LF}_{4,1,-2} [m_1, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{2}{81} g_1^4 \text{LF}_{2,1,0} [m_1, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{2}{81} g_1^4 \text{LF}_{2,2,-1} [m_1, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{4}{81} g_1^4 \text{LF}_{3,1,-1} [m_1, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{2}{81} g_1^4 \text{LF}_{4,1,-2} [m_1, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{24} g_1^2 g_2^2 \text{LF}_{2,1,0} [m_2, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{24} g_1^2 g_2^2 \text{LF}_{2,2,-1} [m_2, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{12} g_1^2 g_2^2 \text{LF}_{3,1,-1} [m_2, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{24} g_1^2 g_2^2 \text{LF}_{4,1,-2} [m_2, m_l^{i1}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{24} g_1^2 g_2^2 \text{LF}_{2,1,0} [m_2, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{24} g_1^2 g_2^2 \text{LF}_{2,2,-1} [m_2, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{1}{12} g_1^2 g_2^2 \text{LF}_{3,1,-1} [m_2, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{24} g_1^2 g_2^2 \text{LF}_{4,1,-2} [m_2, m_l^{i2}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{2}{27} g_1^2 g_3^2 \text{LF}_{2,1,0} [m_3, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{2}{27} g_1^2 g_3^2 \text{LF}_{2,2,-1} [m_3, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{4}{27} g_1^2 g_3^2 \text{LF}_{3,1,-1} [m_3, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{2}{27} g_1^2 g_3^2 \text{LF}_{4,1,-2} [m_3, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{2}{27} g_1^2 g_3^2 \text{LF}_{2,1,0} [m_3, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{2}{27} g_1^2 g_3^2 \text{LF}_{2,2,-1} [m_3, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{4}{27} g_1^2 g_3^2 \text{LF}_{3,1,-1} [m_3, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{2}{27} g_1^2 g_3^2 \text{LF}_{4,1,-2} [m_3, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{2}{9} g_1^2 \overline{y_e}^{i2p} y_e^{i1p} \text{LF}_{2,1,0} [m_e^p, \tilde{\mu}] \delta_{i3i4} - \frac{1}{9} g_1^2 \overline{y_e}^{i2p} y_e^{i1p} \text{LF}_{2,2,-1} [m_e^p, \tilde{\mu}] \delta_{i3i4} - \\ & \frac{1}{9} g_1^2 \overline{y_e}^{i2p} y_e^{i1p} \text{LF}_{3,1,-1} [m_e^p, \tilde{\mu}] \delta_{i3i4} + \frac{1}{36} g_1^4 \text{LF}_{2,1,0} [m_l^{i1}, m_1] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{72} g_1^4 \text{LF}_{3,1,-1} [m_l^{i1}, m_1] \delta_{i1i2} \delta_{i3i4} + \frac{1}{12} g_1^2 g_2^2 \text{LF}_{2,1,0} [m_l^{i1}, m_2] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{24} g_1^2 g_2^2 \text{LF}_{3,1,-1} [m_l^{i1}, m_2] \delta_{i1i2} \delta_{i3i4} + \frac{1}{36} g_1^4 \text{LF}_{2,1,0} [m_l^{i2}, m_1] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{72} g_1^4 \text{LF}_{3,1,-1} [m_l^{i2}, m_1] \delta_{i1i2} \delta_{i3i4} + \frac{1}{12} g_1^2 g_2^2 \text{LF}_{2,1,0} [m_l^{i2}, m_2] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{24} g_1^2 g_2^2 \text{LF}_{3,1,-1} [m_l^{i2}, m_2] \delta_{i1i2} \delta_{i3i4} + \frac{1}{18} g_1^2 \overline{y_u}^{p13} y_u^{p14} \text{LF}_{2,1,0} [m_q^p, \tilde{\mu}] \delta_{i1i2} - \\ & \frac{1}{36} g_1^2 \overline{y_u}^{p13} y_u^{p14} \text{LF}_{2,2,-1} [m_q^p, \tilde{\mu}] \delta_{i1i2} - \frac{1}{36} g_1^2 \overline{y_u}^{p13} y_u^{p14} \text{LF}_{3,1,-1} [m_q^p, \tilde{\mu}] \delta_{i1i2} + \\ & \frac{4}{81} g_1^4 \text{LF}_{2,1,0} [m_u^{i3}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{2}{81} g_1^4 \text{LF}_{3,1,-1} [m_u^{i3}, m_1] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{4}{27} g_1^2 g_3^2 \text{LF}_{2,1,0} [m_u^{i3}, m_3] \delta_{i1i2} \delta_{i3i4} - \frac{2}{27} g_1^2 g_3^2 \text{LF}_{3,1,-1} [m_u^{i3}, m_3] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{4}{81} g_1^4 \text{LF}_{2,1,0} [m_u^{i4}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{2}{81} g_1^4 \text{LF}_{3,1,-1} [m_u^{i4}, m_1] \delta_{i1i2} \delta_{i3i4} + \\ & \frac{4}{27} g_1^2 g_3^2 \text{LF}_{2,1,0} [m_u^{i4}, m_3] \delta_{i1i2} \delta_{i3i4} - \frac{2}{27} g_1^2 g_3^2 \text{LF}_{3,1,-1} [m_u^{i4}, m_3] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{9} g_1^2 \overline{y_e}^{i2p} y_e^{i1p} \text{LF}_{2,1,0} [\tilde{\mu}, m_e^p] \delta_{i3i4} + \frac{1}{18} g_1^2 \overline{y_e}^{i2p} y_e^{i1p} \text{LF}_{3,1,-1} [\tilde{\mu}, m_e^p] \delta_{i3i4} - \\ & \frac{1}{18} g_1^2 \overline{y_e}^{i2p} y_e^{i1p} \text{LF}_{4,1,-2} [\tilde{\mu}, m_e^p] \delta_{i3i4} - \frac{1}{36} g_1^2 \overline{y_u}^{p13} y_u^{p14} \text{LF}_{2,1,0} [\tilde{\mu}, m_q^p] \delta_{i1i2} + \\ & \frac{11}{36} g_1^2 \overline{y_u}^{p13} y_u^{p14} \text{LF}_{3,1,-1} [\tilde{\mu}, m_q^p] \delta_{i1i2} - \frac{1}{9} g_1^2 \overline{y_u}^{p13} y_u^{p14} \text{LF}_{4,1,-2} [\tilde{\mu}, m_q^p] \delta_{i1i2} - \\ & \frac{1}{18} g_1^4 \text{LF}_{2,1,1,-1} [m_1, m_l^{i1}, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{9} g_1^4 m_1^2 \text{LF}_{2,1,1,0} [m_1, m_l^{i1}, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \frac{1}{18} g_1^4 \text{LF}_{2,1,1,-1} [m_1, m_l^{i2}, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{9} g_1^4 m_1^2 \text{LF}_{2,1,1,0} [m_1, m_l^{i2}, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \left. \frac{1}{2} \tilde{\mu}^2 \overline{y_e}^{i2p} y_e^{i1p} \overline{y_u}^{ri3} y_u^{ri4} \text{LF}_{2,1,1,0} [\tilde{\mu}, m_e^p, m_q^r] \right) \end{aligned}$$